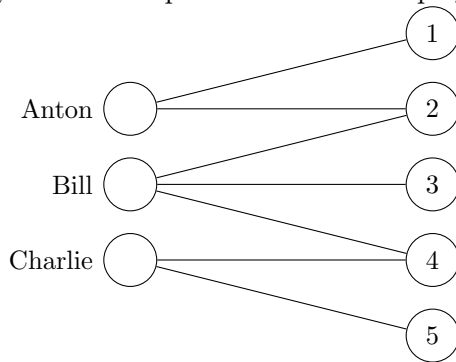


Group work problem solving session 2

September 28, 2026

Problem 1: Integer Programming and Graphs

Alison, Bill and Charlie need to divide up 5 tasks. Each person has a subset of the tasks which they are able to perform. This is displayed in the following graph:



1. They want to divide things up so that the maximum difference between the number of tasks any two people need to perform is as small as possible. Model this question using an integer linear program. If it is easier, you can imagine that instead of this particular bipartite graph, you are working with a general bipartite graph. Hint: introduce a variable t that represents the gap between any two people of how much work they need to do. What linear inequalities does t need to satisfy?
2. For this particular graph, suppose that we use linear programming instead of an integer programming (that is, we drop the constraint that the feasible solutions are integer vectors). How could that affect the outcome of this question?

Problem 2: Graph Drawing and NLP

A group of N people are in a circular room of radius 1. They want to spread out as much as possible, so that for each one of them, the minimum distance between any two people is as large as possible

1. Formulate a nonlinear program that captures this situation.

2. Suppose that in addition, some of these people are friendly with each other, and that this is represented by a graph G on N vertices, so that two people are joined by an edge if they are friends. Two people who are friends with each other are also required to stay within a distance of ϵ with each other. How can we modify the nonlinear program in part (a) to model this additional requirement.
3. It is often desirable to draw a picture of a graph on a screen, so that overall the vertices are spread apart so they are easier to see, but vertices connected by an edge are close together to represent the graph better. The previous question is an optimization problem that gives one way of balancing these two goals. Try to formulate another nonlinear program that can be used to draw a graph.

Problem 3: LP Outcomes

Susan wants to plan what to eat for lunch. Susan wants to make sure to get enough of 3 nutrients: vitamin A, vitamin B and vitamin C, and she has 3 food options available: Carrots, Peas and Potatoes. She lets x_1, x_2 , and x_3 be the amount of Carrots, Peas, and Potatoes respectively, and she wants to maximize how much she enjoys the food she eats. This amount of enjoyment is given by the linear function $x_1 + 2x_2 + 3x_3$. She formulates linear program for this problem

$$\begin{array}{ll} \max & x_1 + 2x_2 + 3x_3 \\ \text{s.t.} & Ax \geq b \\ & x \geq 0. \end{array}$$

Here, A is the matrix representing the how much of each vitamin the different food items possess.

$$A = \begin{pmatrix} 1 & 2 & 1 \\ 2 & 1 & 2 \\ 2 & 1 & 0 \end{pmatrix},$$

and b is a vector representing how much of each nutrient Susan needs to eat.

$$b = \begin{pmatrix} 4 \\ 2 \\ 2 \end{pmatrix}$$

1. Explain why this linear program is a bad model for what Susan wants. Justify this explanation. (Hint: think about which of the possible outcomes of an LP this LP will obtain.)
2. Susan looked at the LP and realized that there was an issue, and tried to correct it by adding a constraint saying that the total amount of food she eats $x_1 + x_2 + x_3$ should be at most 1. There is a different problem with this LP, what is it? Justify this explanation.
3. Susan again adjusts the LP she is solving; instead of requiring that the total amount of food is at most 1, she requires that it is at most 2. What is an optimal solution to this LP? (Hint: at one optimal point, two of the entries are 0).

Solutions

Solution to Problem 1

Part 1

For a general bipartite graph G with vertex set $V = V_1 \cup V_2$, and edges E , we create a variable x_{ij} for each edge $\{i, j\}$ to indicate that person i takes on task j . We want x_{ij} to be 1 if task j is assigned to person i and 0 otherwise.

The number of tasks that person i takes on is then

$$t_i = \sum_{j: \{i, j\} \in E} x_{ij}.$$

We want to minimize the maximum difference between t_i and $t_{i'}$ for persons i and i' . The maximum function is not linear, so we cannot represent this directly. Let's introduce a new variable T which represents the maximum such difference. For each $i \neq i'$, we know then that

$$T \geq t_i - t_{i'}$$

and also that

$$T \geq t_{i'} - t_i.$$

Note that if we just include these inequalities for possible ordered pairs (i', i) where $i, i' \in V$, then this is the same.

If we then minimize T subject to these constraints, then T must in fact equal the maximum such difference (it cannot be smaller because of the constraints and if it were larger, then we could get a smaller objective by just decreasing T to be the maximum).

Thus, the overall integer program is

$$\begin{array}{ll} \max & T \\ \text{s.t.} & \forall i, i' \in V, T \geq t_i - t_{i'} \\ & \forall i \in V, t_i = \sum_{j: \{i, j\} \in E} x_{ij} \\ & 0 \leq x_{ij} \leq 1 \\ & x_{ij} \in \mathbb{Z}. \end{array}$$

For this particular graph, we would have 7 variables x_{ij} , one for each edge, and 6 inequality

constraints, one for each pair of people.

$$\begin{array}{ll}
\max & T \\
\text{s.t.} & T \geq t_1 - t_2 \\
& T \geq t_1 - t_3 \\
& T \geq t_2 - t_3 \\
& T \geq t_2 - t_1 \\
& T \geq t_3 - t_1 \\
& T \geq t_3 - t_2 \\
& t_1 = x_{11} + x_{12} \\
& t_2 = x_{22} + x_{23} + x_{24} \\
& t_3 = x_{34} + x_{35} \\
& x_{11}, x_{12}, x_{22}, x_{23}, x_{24}, x_{34}, x_{35} \in \mathbb{Z} \\
& 0 \leq x_{11}, x_{12}, x_{22}, x_{23}, x_{24}, x_{34}, x_{35} \leq 1
\end{array}$$

Part 2

If we are allowed to assign people a fraction of a task instead of a whole task, then we can give everyone equal work: give Anton task 1 and 2/3 of task 2. Give Bill 1/3 of task 2, all of task 3, and 1/3 of task 4. Finally give Charlie 2/3 of task 4 and all of task 5. Notice how everyone has 5/3 of a task this way, and every task has a total of 1 person assigned to it.

Solution to Problem 2

Part 1

$$\begin{array}{ll}
\max & \min_{i \neq j} \|x_i - x_j\| \\
\text{s.t.} & \|x_i\| \leq 1 \\
& x_i \in \mathbb{R}^2.
\end{array}$$

Part 2

$$\begin{array}{ll}
\max & \min_{i \neq j} \|x_i - x_j\| \\
\text{s.t.} & \|x_i\| \leq 1 \\
& \forall i, j \in E(G), \|x_i - x_j\| \leq \epsilon \\
& x_i \in \mathbb{R}^2.
\end{array}$$

Part 3

This is intentionally open ended. For example, you could instead try to minimize the distances between vertices paired by an edge, subject to constraints that all pairs of vertices not paired by an edge must be distance at least 1 apart from each other.

Solution to Problem 3

Part 1

In this case, the model is bad because the linear program is unbounded; we can see this because if we increase any entry of x , we can make the objective larger while keeping the constraints valid. We can also say that for $(1, 0, 0)$ is a recession direction for this linear program.

Part 2

Adding this constraint will make the linear program infeasible. To see this, note that adding the constraints together gives

$$\begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} \begin{pmatrix} 1 & 2 & 1 \\ 2 & 1 & 2 \\ 2 & 1 & 0 \end{pmatrix} x = \begin{pmatrix} 5 \\ 5 \\ 5 \end{pmatrix}^T x \geq \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} b = 8,$$

so $\begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} x \geq \frac{8}{5} > 1$, so no feasible point can satisfy this constraint.

Part 3

The optimal solution is $(0, 2, 0)$. First note that this is feasible; the sum of the entries is 2, each entry is nonnegative, and

$$\begin{pmatrix} 1 & 2 & 1 \\ 2 & 1 & 2 \\ 2 & 1 & 0 \end{pmatrix} \begin{pmatrix} 0 \\ 2 \\ 0 \end{pmatrix} = \begin{pmatrix} 4 \\ 2 \\ 2 \end{pmatrix} \geq \begin{pmatrix} 4 \\ 2 \\ 2 \end{pmatrix}.$$

We can also check optimality by noting that $x_1, x_2, x_3 \geq 0$, we can see that

$$x_1 + 2x_2 + 3x_3 \leq 2x_1 + 2x_2 + 3x_3 = 4(x_1 + x_2 + x_3) - (x_1 + 2x_2 + 2x_3) = 4(x_1 + x_2 + x_3) - \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}^T Ax$$

So for any feasible point, where $Ax \geq \begin{pmatrix} 4 \\ 2 \\ 2 \end{pmatrix}$ and $x_1 + x_2 + x_3 = 2$, the objective satisfies

$$x_1 + 2x_2 + 3x_3 \leq 8 - 4 = 4.$$